

(Supplementary Table 2). They include three observational SST reconstructions: the Hadley Centre Sea Ice and Sea Surface Temperature dataset v.1.1 (ref. 81), Extended Reconstructed Sea Surface Temperature v.5 (ref. 82) and the Centennial in situ Observation-Based Estimates of Sea Surface Temperature v.2 (ref. 83) for 1871–2023, and six reanalysed SST and ocean temperature datasets: GECCO3 for 1950–2018 (the German contribution to Estimating the Circulation and Climate of the Ocean v.3)⁸⁴, Global Ocean Data Assimilation System for 1980–2023 (ref. 85), the ECMWF Ocean Reanalysis System 5 for 1958–2023 (ref. 86), ORA20C for 1900–2009 (ensemble of ten-member ECMWF Ocean Reanalysis of the 20th Century)⁸⁷, the Predictive Ocean Atmosphere Model for Australia Ensemble Ocean Data Assimilation System for 1960–2014 (ref. 88) and Simple Ocean Data Assimilation Phase v.2.2.4 for 1871–2010 (ref. 89). The thermocline depth is defined as the depth of the 20 °C isotherm. We also use surface air temperature from the ERA5 reanalysis⁹⁰, and gridded precipitation from the Climate Prediction Center Merged Analysis of Precipitation⁹¹ for 1979–2022. The monthly anomaly fields were calculated by removing the monthly climatology for the period of 1979–2022 and the quadratic trend over the whole period. We have focused on the satellite era from 1979 onwards because SST observations are sparse in the presatellite period.

Climate forecast and hindcast data

We use the 3 month averaged Niño3.4 index forecasts from the operational IRI ENSO Forecast product⁵. We also use SST hindcasts and real-time forecasts from ten models participating in the NMME project⁹². The ensemble sizes range from 10 to 24 for each model (Supplementary Table 4). The monthly forecast anomalies were calculated with respect to the monthly climatology from January 1982 to December 2010 for each member and forecast lead. For CCSM4 and CFSv2, we eliminate the discontinuous forecast biases by calculating the forecast anomalies using two different climatological periods of 1982–1998 and 1999–2010, respectively, following ref. 45

In addition, we use the Niño3.4, Niño3 and Niño4 indices forecasts from an AI model (the 3D-Geformer ENSO neural network model⁹) covering the period of 1983–2021. This model demonstrated ENSO forecast skills comparable with the convolutional neural networks model developed by Ham et al.⁸, which is among the most skilful AI ENSO forecasts^{93,94}.

Stochastically forced XRO simulations

To assess the XRO's performance in simulating ENSO and mode interactions, we conducted stochastically forced simulations using the operators and stochastic forcing matrices estimated from the ORAS5 reanalysis for 1979–2022 (black curves in Extended Data Fig. 1). We numerically integrate equations (1) and (2) with a time step of 0.01 month for 45,000 years and archive monthly averaged states for the analysis. The last 43,000 years were analysed and split into 1,000 non-overlapping epochs of 43 years each, aligning with the observational record length. An example of a simulated Niño3.4 SSTA index for the ten consecutive centuries is shown in Supplementary Fig. 3.

Out-of-sample reforecasts

To perform robust out-of-sample testing of the XRO performance, we next use observational data including the presatellite period because at least 40–50 years of data are required to get a robust XRO fit. We choose to discard data before 1950 as there are large uncertainties in the SSTA and equatorial thermocline depth indices (Supplementary Fig. 15). Therefore, we fitted the XRO and nRO models on 1950–1999 (50 years) data, conducted deterministic retrospective 21 month forecasts by integrating the XRO (equation (1)) and nRO (equation (6)) initialized from observed state values for the period of 2002–2022, and verified the model against observations in the 2002–2022 period. To access the impact of the decadal change in the performance of the

XRO in forecasting ENSO, we also verified the model on two other 21 year non-overlapping periods: the previous period 1950–1970 (in which period of 1973–2022 data was used for training) and the middle period 1972–1992 (in which the periods of 1950–1970 and 1994–2022 data was used for training). The multi-data-products ensemble mean SSTA and WWV anomaly indices were used for fitting and verification.

Control XRO and nRO reforecasts

Using the operator and stochastic forcing parameters estimated from the ORAS5 reanalysis for 1979–2022, we conducted a control experiment by integrating the XRO (equation (1)) initialized from observed state values of $[T_{\text{ENSO}}, h, T_{\text{NPM}}, T_{\text{SPMM}}, T_{\text{IOB}}, T_{\text{IOD}}, T_{\text{SIOD}}, T_{\text{TNA}}, T_{\text{ATL3}}, T_{\text{SASD}}]$ with retrospective 21 month forecasts for the period of January 1979–October 2023 (refer to XRO). The ensemble mean forecast of 100 members is almost identical to the deterministic forecast in which the stochastic forcing terms are neglected during the integration (Supplementary Fig. 16a,b). Although the 100-member stochastic XRO forecasts provide an opportunity for probabilistic ENSO forecasts (Supplementary Fig. 16c–f), here we focus on the deterministic skill and neglect the stochastic forcing terms in all the remaining forecast experiments. Similarly, we conducted a nRO deterministic experiment by integrating equation (6) initialized from observed state values of $[T_{\text{ENSO}}, h]$.

Cross-validated reforecasts

We carried out cross-validated forecasts using both the XRO and nRO models from the ORAS5 reanalysis for 1979–2022 using a jackknife subsampling approach. We sequentially excluded 3 year segments of data (1979–1981, 1982–1985, 1986–1989, 1990–1993, 1994–1997, 1998–2001, 2002–2005, 2006–2009, 2010–2013, 2014–2017, 2018–2021 and 2022), then trained the model operator parameters based on the remaining data. Subsequently, we generated forecasts for each month during the years not included in the model fitting. The uncertainty in the fitted parameters is illustrated as black shading in Extended Data Fig. 1. The skill of cross-validated forecast is not sensitive to the choice of excluding from 2 to 7 years (Supplementary Fig. 8a).

Large ensemble simulations and perfect model reforecasting experiments

To assess of the robustness of the XRO fitting and forecasting performance, we use large ensemble (LENS) historical simulations for four climate models: Community Earth System Model v.1 (CESM1)⁹⁵, v.2 (CESM2)⁹⁶, Model for Interdisciplinary Research on Climate v.6 (MIROC6)⁹⁷ and the Max Planck Institute for Meteorology Earth System Model v.1.1 (MPI-ESM)⁹⁸. Each LENS was generated by repeatedly running the same model simulation with identical external forcing but with small initial condition differences. The number of members for each LENS used in this study are as follows: 39 for CESM1, 100 for CESM2, 50 for MIROC6 and 99 for MPI-ESM. We use the historical period of 1959–2002, aligning it with the observational record length (43 years).

We performed the 'perfect model' reforecast in which the XRO model was trained by the LENS output and tasked to reforecast itself instead of the observations. We carried out twin experiments for each LENS (Supplementary Fig. 8b–e): the same-member reforecast experiment, in which the XRO model is repeatedly fitted for a member, forecasted and verified against the same member. This aligns with the XRO control experiment for the observations. In the cross-member reforecast experiment, the XRO model is fitted for a specific member but forecasted and verified against a different member (an independent realization in the LENS). Specifically, we forecast ensemble member j using the two versions of XRO models, which were fitted on member j_1 and j_2 data, respectively, and repeated the process for all members within the LENS. The skill difference between the cross-member experiment